Introduction

Project Title: Customer Purchase Amount Prediction using Machine Learning

Objective

The main objective of this project is to build a predictive model that estimates the purchase amount of customers based on their demographic and behavioral attributes such as:

Age
Experience
Income
Education
Region

This helps businesses in:

Targeted marketing
Customer segmentation
Revenue forecasting

2. Dataset Description

The dataset customer_sales.csv contains customer-related information.

Features

Feature Name	Description
Age	Age of customer
Experience	Years of experience
Income	Annual income
Education_years	Years of education
Region	Customer location (categorical)
Purchase_amount	Target variable

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3. Data Exploration & Analysis

- Initial Checks
- Dataset shape and structure examined
- First few rows displayed
- Data types verified
- Missing Values
- No missing values found

Visualization:

- Histogram used to analyze distribution of purchase_amount
- Helps understand:
- Skewness
- Spread of data

4. Data Preprocessing

- Steps Performed
- Copy dataset for safe processing
- Categorical Encoding
- Region converted into numeric format
- Feature Selection
- Target: purchase_amount
- Features: Remaining columns
- Correlation Analysis
- Identified which features strongly influence purchase amount

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5. Train-Test Split

Data split into:

- Training set (80%)
- Testing set (20%)
- Ensures:
- Model generalization
- Fair performance evaluation

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6. Model Building

Two machine learning models were implemented:

Model 1: Linear Regression

- Simple and interpretable model
- Assumes linear relationship

Model 2: Random Forest Regressor

- Ensemble learning method
- Handles non-linearity better
- More robust and accurate

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7. Model Evaluation

Metrics Used:

- MAE (Mean Absolute Error) → average error
- RMSE (Root Mean Square Error) → penalizes large errors
- R^2 Score → model accuracy

Visualization:

- Actual vs Predicted plots
- Residual plots for error analysis

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8. Model Comparison

Metric	Linear Regression	Random Forest
MAE	Lower/Moderate	Lower
RMSE	Moderate	Lower
R^2	Good	Better ✅

Conclusion:
Random Forest performs better than Linear Regression.

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9. Prediction on New Data

Model used to predict purchase amount for new customers:

Example Input:
Age: 30, 45, 55
Income: 50K, 72K, 95K
Region: North, South, West

Output:
Predicted purchase amounts generated successfully

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10. Key Findings

- Income has strong impact on purchase amount
- Experience and education also influence spending
- Random Forest captures complex patterns better
- Linear Regression is useful as a baseline

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11. Conclusion

- Successfully built a predictive model
- Compared multiple algorithms
- Achieved better accuracy using Random Forest

Business Impact:

- Helps in customer targeting
- Improves marketing strategy
- Increases profitability

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12. Future Scope

Use larger datasets
Apply advanced models (XGBoost, Neural Networks)
Deploy model as web application
Add real-time prediction system

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13. Technologies Used

- Python
- Pandas, NumPy
- Matplotlib, Seaborn
- Scikit-learn